

ACTIVE DIRECTORY

- A directory is a hierarchical structure that stores information about objects on the network. A directory service, such as Active Directory Domain Services (AD DS), provides the methods for storing directory data and making this data available to network users and administrators. For example, AD DS stores information about user accounts, such as names, passwords, phone numbers, and so on, and enables other authorized users on the same network to access this information.
- Active Directory stores information about objects on the network and makes this information easy for administrators and users to find and use. Active Directory uses a structured data store as the basis for a logical, hierarchical organization of directory information.
- This data store, also known as the directory, contains information about Active Directory objects. These objects typically include shared resources such as servers, volumes, printers, and the network user and computer accounts.
- Security is integrated with Active Directory through logon authentication and access control to objects in the directory. With a single network logon, administrators can manage directory data and organization throughout their network, and authorized network users can access resources anywhere on the network. Policy-based administration eases the management of even the most complex network.

ACTIVE DIRECTORY

AD DS helps manage network operations by providing a structured way to store data in a hierarchical organization. This makes it easier for administrators to manage user access rights and system configurations across different domains within the same network. AD DS also integrates security by authenticating login functions and controlling access to directory resources. It does this through:

User authentication. AD DS authenticates users before they can access resources on the network, ensuring only authorized individuals have entry to specific parts of the system.

Data storage. It stores directory data, like usernames, passwords and phone numbers, which help streamline operations within an organization.

Policies enforcement. With group policy objects (GPO), administrators can enforce security policies across multiple machines at once, saving time while maintaining high levels of security.

ACTIVE DIRECTORY

Active Directory also includes:

- A set of rules, the schema, that defines the classes of objects and attributes contained in the directory, the constraints and limits on instances of these objects, and the format of their names.
- A query and index mechanism, so that objects and their properties can be published and found by network users or applications.
- A replication service that distributes directory data across a network. All domain controllers in a domain participate in replication and contain a complete copy of all directory information for their domain. Any change to directory data is replicated to all domain controllers in the domain.

Understanding AD

Administrators use Active Directory to store and organize objects on a network (such as users, computers, devices, and so on) into a secure hierarchical containment structure that is known as the logical structure. Although the logical structure of Active Directory is a hierarchical organization of all users, computers, and other physical resources, the forest and domain form the basis of the logical structure. Forests, which are the security boundaries of the logical structure, can be structured to provide data and service autonomy and isolation in an organization in ways that can both reflect site and group identities and remove dependencies on the physical topology.

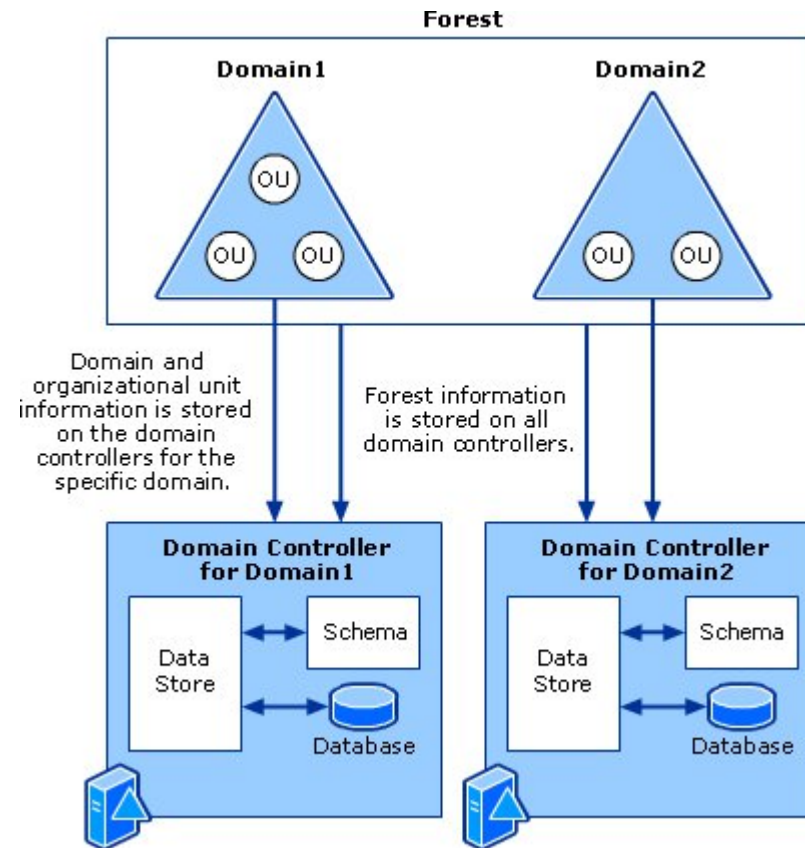
Understanding AD

Active Directory Structure and Storage Technologies consists of four parts

- Active Directory domains and forests. Forests, domains, and organizational units (OUs) make up the core elements of the Active Directory logical structure. A forest defines a single directory and represents a security boundary. Forests contain domains.
- Domain Name System (DNS) support for Active Directory. DNS provides a name resolution service for domain controller location and a hierarchical design that Active Directory can use to provide a naming convention that can reflect organizational structure.
- Schema. The schema provides object definitions that are used to create the objects that are stored in the directory.
- Data store. The data store is the portion of the directory that manages the storage and retrieval of data on each domain controller.

Understanding AD

The following figure illustrates the Active Directory data structure and storage architecture.



DNS Support for AD

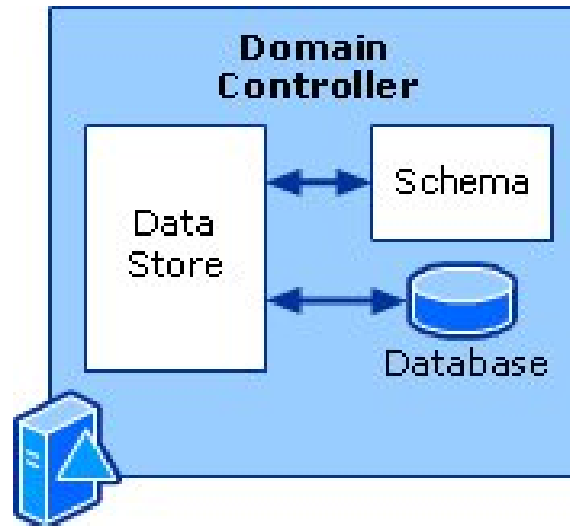
Active Directory uses DNS as its domain controller location mechanism. When any of the principal Active Directory operations, such as authentication, updating, or searching, is performed, domain joined computers use DNS to locate Active Directory domain controllers, and these domain controllers use DNS to locate each other. For example, when a network user with an Active Directory user account logs on to an Active Directory domain, the user's computer uses DNS to locate a domain controller for the Active Directory domain to which the user wants to log on.

To log on to a network that consists of an Active Directory forest, a client workstation must first be able to locate a nearby domain controller. The domain controller is necessary for initial authentication of both the workstation and the user and for subsequent authorization of the user for the files and resources to which the user needs access. The support that is provided to Active Directory by DNS enables a client workstation to locate a domain controller.

AD Schema

The Active Directory schema contains definitions for all the objects that are used to store information in the directory. There is one schema per forest. However, a copy of the schema exists on every domain controller in the forest. This way, every domain controller has quick access to any object definition that it might need, and every domain controller uses the same definition when it creates a given object. The data store relies on the schema to provide object definitions, and the data store uses those definitions to enforce data integrity. The result is that all objects are created uniformly, and it does not matter which domain controller creates or modifies an object because all domain controllers use the same object definition.

The following figure illustrates the relationship of the schema to the data store in the schema architecture.



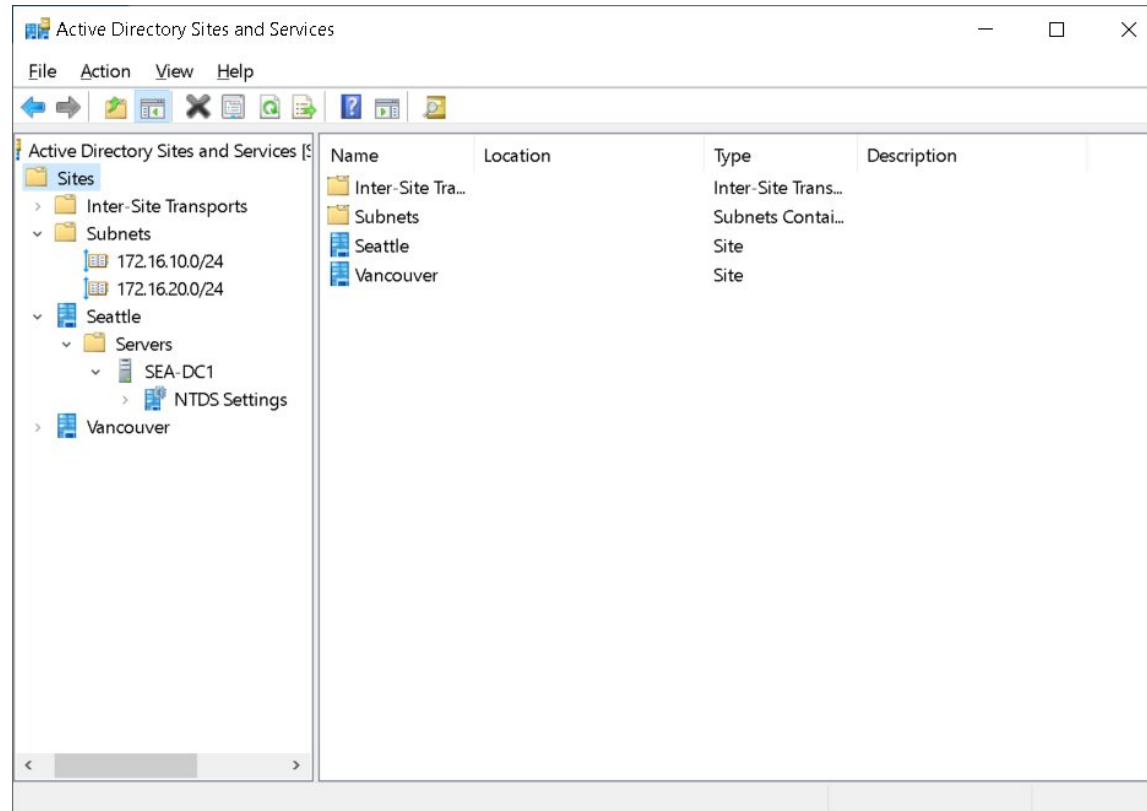
The logical Structure of AD

AD DS logical components are structures that you use to implement an AD DS design that is appropriate for an organization. The following table describes the types of logical components that an AD DS database contains.

Logical Component	Description
Schema	A schema is the set of definitions of the object types and attributes that you use to define the objects created in AD DS.
Domain	A domain is a logical administrative container for objects such as users and computers. A domain maps to a specific partition and you can organize the domain with parent-child relationships to other domains.
Domain tree	A domain tree is a hierarchical collection of domains that share a common root domain and a contiguous Domain Name System (DNS) namespace.
Forest	A forest is a collection of one or more domains that have a common AD DS root, a common schema, and a common global catalog.
OU	An OU is a container object for users, groups, and computers that provides a framework for delegating administrative rights and administration by linking Group Policy Objects (GPOs).

The physical Contents of AD

Physical components in AD DS are those objects that are tangible, or that described tangible components in the real world.



The physical Contents of AD

The following table describes some of the physical components of AD DS.

Physical component	Description
Domain controller	A domain controller contains a copy of the AD DS database. For most operations, each domain controller can process changes and replicate the changes to all the other domain controllers in the domain.
Data store	A copy of the data store exists on each domain controller. The AD DS database uses Microsoft Jet database technology and stores the directory information in the Ntds.dit file and associated log files. The C:\Windows\NTDS folder stores these files by default.
Global catalog server	A global catalog server is a domain controller that hosts the global catalog, which is a partial, read-only copy of all the objects in a multiple-domain forest. A global catalog speeds up searches for objects that might be stored on domain controllers in a different domain in the forest.
Read-only domain controller (RODC)	An RODC is a special read only installation of AD DS. RODCs are common in branch offices where physical security isn't optimal, IT support is less advanced than in the main corporate centers, or line-of-business applications need to run on a domain controller.
Site	A site is a container for AD DS objects, such as computers and services that are specific to a physical location. This is in comparison to a domain, which represents the logical structure of objects, such as users and groups, in addition to computers.
Subnet	A subnet is a portion of the network IP addresses of an organization assigned to computers in a site. A site can have more than one subnet.

Domains and Forests (In Detail)

An AD DS forest is a collection of one or more AD DS trees that contain one or more AD DS domains. Domains in a forest share:

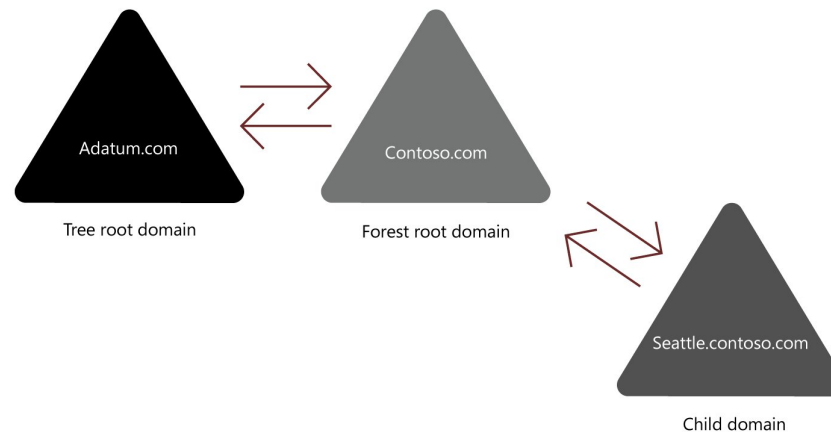
- A common root.
- A common schema.
- A global catalog.

An AD DS domain is a logical administrative container for objects such as:

- Users
- Groups
- Computers

What is AD DS forest

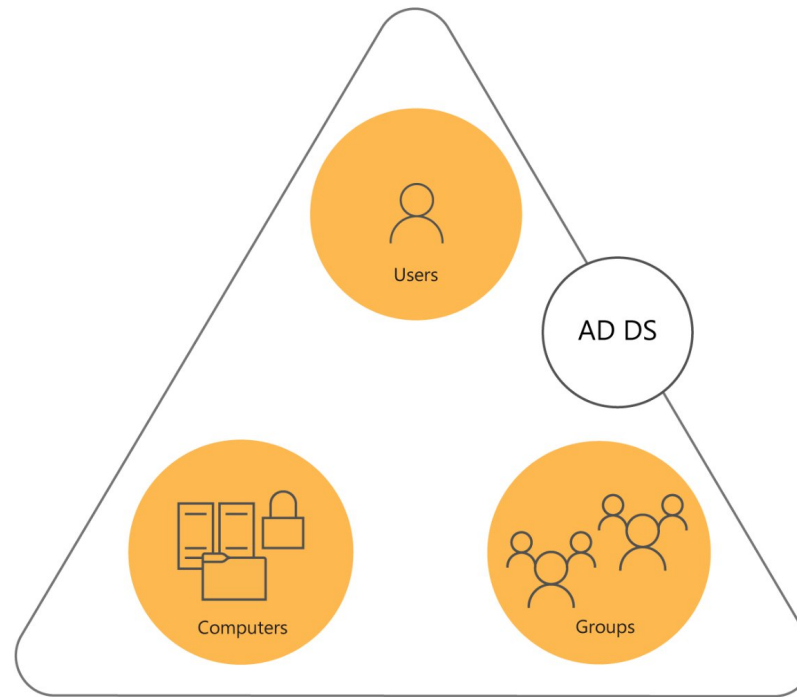
- A forest is a top-level container in AD DS. Each forest is a collection of one or more domain trees that share a common directory schema and a global catalog. A domain tree is a collection of one or more domains that share a contiguous namespace. The forest root domain is the first domain that you create in the forest.
- The forest root domain contains objects that don't exist in other domains in the forest. Because you always create these objects on the first domain controller, a forest can consist of as few as one domain with a single domain controller, or it can consist of several domains across multiple domain trees.
- The following graphic displays Contoso.com as the forest root domain. Beneath are two domains, Adatum.com in a separate tree, and Seattle.Contoso.com as a child of Contoso.com.



What is AD DS domain

An AD DS domain is a logical container for managing user, computer, group, and other objects. The AD DS database stores all domain objects, and each domain controller stores a copy of the database.

The following graphic displays an AD DS domain. It contains users, computers, and groups.



Commonly used objects in Domain

Object	Description
User accounts	User accounts contain information about users, including the information required to authenticate a user during the sign-in process and build the user's access token.
Computer accounts	Each domain-joined computer has an account in AD DS. You can use computer accounts for domain-joined computers in the same way that you use user accounts for users.
Groups	Groups organize users or computers to simplify the management of permissions and Group Policy Objects in the domain.

An AD DS domain provides

An AD DS domain provides:

- Authentication. Whenever a domain-joined computer starts or a user signs in to a domain-joined computer, AD DS authenticates it. Authentication verifies that computer or user has the proper identity in AD DS by verifying its credentials.
- Authorization. Windows uses authorization and access control technologies to determine whether to allow authenticated users to access resources.

What are trust relationships?

An AD DS trusts enable access to resources in a complex AD DS environment. When you deploy a single domain, you can easily grant access to resources within the domain to users and groups from the domain. When you implement multiple domains or forests, you should ensure that the appropriate trusts are in place to enable the same access to resources.

In a multiple-domain AD DS forest, two-way transitive trust relationships generate automatically between AD DS domains so that a path of trust exists between all the AD DS domains.

When you set up trusts between domains within the same forest, across forests, or with an external realm, Windows Server creates a trusted domain object to store the trusts' information, such as transitivity and type, in AD DS. Windows Server stores this trusted domain object in the System container in AD DS.

Main trust types

Trust type	Description	Direction	Description
Parent and child	Transitive	Two-way	When you add a new AD DS domain to an existing AD DS tree, you create new parent and child trusts.
Tree-root	Transitive	Two-way	When you create a new AD DS tree in an existing AD DS forest, you automatically create a new tree-root trust.
External	Nontransitive	One-way or two-way	External trusts enable resource access with a Windows NT 4.0 domain or an AD DS domain in another forest. You also can set these up to provide a framework for a migration.
Realm	Transitive or nontransitive	One-way or two-way	Realm trusts establish an authentication path between a Windows Server AD DS domain and a Kerberos version 5 (v5) protocol realm that implements by using a directory service other than AD DS.
Forest (complete or selective)	Transitive	One-way or two-way	Trusts between AD DS forests allow two forests to share resources.
Shortcut	Nontransitive	One-way or two-way	Configure shortcut trusts to reduce the time taken to authenticate between AD DS domains that are in different parts of an AD DS forest. No shortcut trusts exist by default, and an administrator must create them if they're required.

What are OUs

An OU is a container object within a domain that you can use to consolidate users, computers, groups, and other objects. You can link Group Policy Objects (GPOs) directly to an OU to manage the users and computers contained in the OU. You can also assign an OU manager and associate a COM+ partition with an OU.

You can create new OUs in AD DS by using:

- Active Directory Administrative Center.
- Active Directory Users and Computers.
- Windows Admin Center.
- Windows PowerShell with the Active Directory PowerShell module.

Demo

1. Installing Active Directory
2. How to create, manage, and delete users and groups in Active Directory

Kerberos Authentication

Kerberos is a secure authentication protocol, and it is most used as the basis for single sign-on and allows information to be transmitted securely over a network.

Kerberos can also be used as effective access control, as only users that are authenticated can access resources on the network.

Basic Authentication with Kerberos

In almost all authentication processes there are three main players. These are:

- The Client or User that is trying to access a resource on the network.
- The Resource that needs to be accessed by the user.
- The Key Distribution Center (KDC) in Windows environments - we know this as the Domain Controller.

Core components of Kerberos Authentication

Kerberos Realm: a logical network, similar to a domain, over which a Kerberos authentication server has the authority to authenticate a user, host or service.

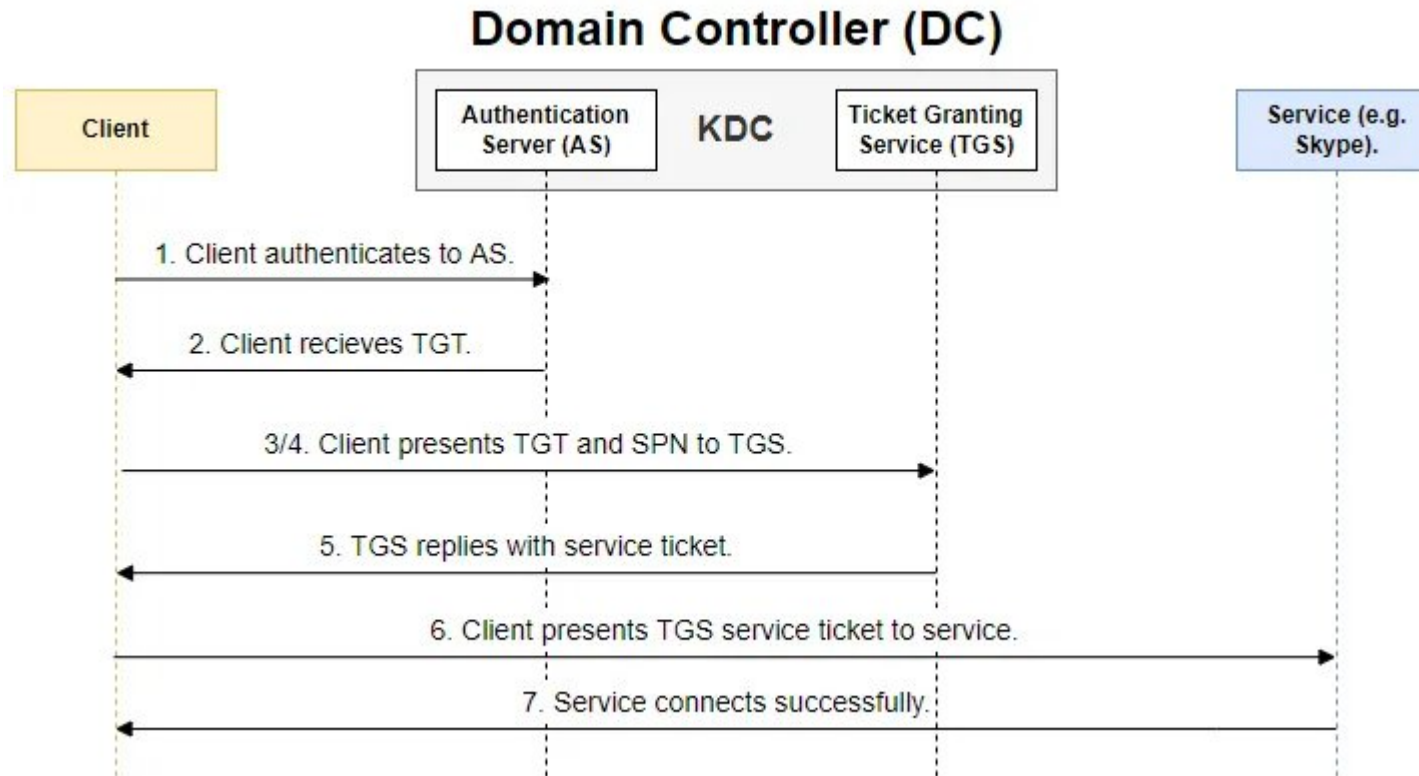
Key Distribution Centre (KDC): contains the Authentication Server (AS) and the Ticket Granting Service (TGS). Its main function is to be a mediator between these two, relaying messages from the AS, grants a ticket-granting ticket (TGT), then passing that to be encrypted by the TGS. The KDC for a domain is located on a domain controller.

Authentication Server (AS): a client authenticates themselves to the AS using a username and password login. The AS then forwards the username to the KDC that in turn grants a TGT.

Ticket Granting Service (TGS): when a client wants to access a service, they must present their TGT to the TGS.

Service Principal Name (SPN): an identifier given to a service instance to associate a service instance with a domain service account.

How Kerberos Authentication works



Thank You